Resolution Geometry: Light as Emission

The Projection Formula, the Mirror as a Quotient, and the Logic of Images

Matěj Rada
MatejRada@email.cz
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Abstract

The round-trip lock showed that the radar echo of a static reflector is a pair of events, an emission and a reception, and that it reconstructs the spatial metric, $B = \frac{c}{2} z T'$. This paper asks whether *all* reflected light can be written that way, as emission and nothing else, and whether such a formula explains the mirror: the image behind the glass, the planet's image "inside" the planet, and the properties of the observer that the image carries back.

We lock a layered *light projection formula*. The received field equals the free propagation of every emitter to the observer, where the emitters are the declared sources together with the induced currents that solve a self-consistent equation. No bounce term exists. Reflection, refraction and the *extinction* of the incident wave inside matter are all emission. This is the Ewald–Oseen theorem, re-derived here and computed from susceptibility alone; it reproduces the Airy slab formulas to second-order convergence.

The formula reduces exactly as follows:

- for a planar mirror, to the charge-conjugate parity (CP) image;
- for a moving mirror, to the Lorentz-reflected image, giving Einstein's 1905 formulas;
- for an arbitrary mirror worldline in 1+1 dimensions, to the null re-projection $g = -f \circ p$;
- for a convex planet seen paraxially, to an image at depth $R/2$ *inside* the planet;
- in static gravity, to the optical-metric round trip, in which your own image is never frequency-shifted.

In RG terms the mirror image is an *equality-of-law quotient point*. It is real as an observable and fictitious as a latent location: behind the mirror the field is exactly zero. The three latent representations, bounce, image and emission, are one CORRESPONDENCE for far-field data. Emission is identified by protocols that probe the reflector: extinction behind it, susceptibility dependence, and finite penetration.

The mirror "reverses" only along its normal. "Left–right" is a choice-dependent attribution; the invariants are chirality and the flip of photon helicity. We list where current physics is complete and where it is open: photon localization, momentum in media, and vacuum emission. For the last, the classical map p is the complete input.

Battery: 11 attacks (10 locked, 1 boundary), 7 computed canaries rejected, master seed 20261025.

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1 From radar to emission

The round-trip battery [1] established a conditional exact theorem in static spherical gravity. With $z = N/N_0$ the clock-rate ratio and $T(r)$ the emitter proper time of an out-and-back radial signal,

$$T(r) = \frac{2N_0}{c} \int_{r_0}^{r} \frac{B}{N}\, ds, \qquad B = \frac{c}{2}\, z\, T'(r).$$

The derivation uses only two *events*: emission at the emitter and re-emission at the reflector. It never uses a "bounce". The question is whether this is a special case of a general law. We claim, and attack, the following thesis.

> *Received light is the projection of emission events onto the observer. A bounce is a shorthand, and an image is a quotient object.*

For the thesis to hold it must explain three things:

- why the line of sight to an image "cuts through" the reflector, while nothing physically passes through;
- why we see ourselves in a mirror, and which of our properties the image carries back;
- how motion and gravity enter.

2 The projection formula

Definition 2.1 (Declarations)**.** Declared are:

- a geometry: Minkowski space, or a static metric whose light propagation is governed by the optical metric h/N^2 [2];
- Maxwell propagation, or its scalar reduction where stated;
- sources J_{src};
- linear media with susceptibility χ, perfect conductors as a limit;
- the observer protocol: the field on the observer's region or worldline.

G denotes the outgoing (retarded) Green operator of the declared geometry.

Theorem 2.2 (Light projection formula)**.** *The received field is*

$$\boxed{E_{\text{obs}} = G\,J_{\text{src}} + G\,J_{\text{ind}}, \qquad J_{\text{ind}} = \mathcal{X}\big(GJ_{\text{src}} + GJ_{\text{ind}}\big),}$$

where $\mathcal{X}$ is the declared material response, which is surface currents for a conductor. Every term is the free propagation of an emitter. There is no reflection operator, and reflection, refraction and extinction all follow from the emission sum.

Proof. This is the integral (Lippmann–Schwinger) form of Maxwell's equations with linear constitutive relations. The induced polarization and current are sources like any other; the Green operator propagates each one freely. For the exactness of the reductions below see Sections 3–6. □

3 Reflection is emission; extinction is emission

Theorem 3.1 (Sheet emission; L01)**.** *At normal incidence on a perfect conductor, the induced current $K = 2\hat{n} \times H_{\text{inc}}$ radiates $-\frac{Z_0}{2} K e^{ik|z|}$ symmetrically. Behind the surface this exactly cancels the incident wave; in front it is exactly the reflected wave.*

Theorem 3.2 (Ewald–Oseen from emission only; L04)**.** *In one dimension, let a slab of susceptibility χ (real, or complex for an absorbing medium) be represented only by self-consistent emitters:*

$$E(z) = e^{ikz} + \frac{ik\chi}{2} \int_{\text{slab}} e^{ik|z-z'|} E(z')\, dz'.$$

Then the emitted reflection and transmission coefficients equal the Airy slab coefficients with $n = \sqrt{1+\chi}$. Inside the slab only $e^{\pm ikz}$ survive: the incident vacuum wave e^{ikz} is extinguished by the emission of the medium itself.

Proof. $\frac{i}{2k} e^{ik|z|}$ is the outgoing Green function of $\partial_z^2 + k^2$. The integral equation is equivalent to $E'' + k^2(1+\chi)E = 0$ in the slab with outgoing conditions outside, whose solution is Airy's. The extinction is the classical theorem of Ewald and Oseen [3, 4, 5]. □

Numerically (Nyström, 400/800/1600 nodes), the errors in r and t converge at order 2.00 to 10^{-6}, for both a glass slab and an absorbing slab. The vacuum-wave coefficient inside is 4×10^{-7}. The first Born approximation, i.e. emission without self-consistency, misses the reflection by 0.47 and 0.35. *This is what distinguishes emission from bounce.* A bounce model takes r and t as inputs; emission *computes* them from χ.

4 The mirror image as a quotient

Theorem 4.1 (CP image; L02)**.** *Let R be the reflection through the mirror plane. For a planar perfect mirror the reflected field is*

$$E_r(x) = -R\,E_i(Rx), \qquad B_r(x) = R\,B_i(Rx).$$

This is charge conjugation composed with parity across the mirror. It solves Maxwell's equations and the boundary conditions $E_{\tan} = 0$, $B_n = 0$ exactly, for oblique TE and TM waves (checked symbolically).

Theorem 4.2 (The image is a quotient point; L03)**.** *Let the observation protocol be the field in the observer's half-space. Then the latent configurations*

$$\textit{A: source + mirror}, \qquad \textit{B: source + CP-image emitter, no mirror}$$

have equal observation laws, so they lie in one fibre and the image is a point of the observable quotient. A protocol behind the surface separates them: A has field 0 there (Theorem 3.1), B does not.

Proof. By Theorem 4.1 the field of B satisfies the boundary-value problem of A on the half-space, and that problem has a unique solution. Battery: 400 random oscillating dipoles and plane points with full near and far fields. The tangential E and normal B vanish to machine zero; behind the plane, configuration B reaches $|E| \approx 15$. □

Remark 4.3 ("Cutting through"). The line of sight to an image passes straight through the reflector, so the image "cuts through" it. That statement belongs to the quotient: the received field is *as if* emitted from behind the surface. The latent field behind a perfect mirror is zero. Light does not pass through; the emission of the surface charges extinguishes it.

Proposition 4.4 (Finite penetration; L10)**.** *In a medium of complex index $n_r + i\kappa$ the intensity decays as $e^{-2\kappa kz}$, with depth $\lambda/(4\pi\kappa)$. For $\kappa = 1$ at 500 nm the depth is 4×10^{-8} m, and transmission through an Earth radius is $10^{-6.95\times10^{13}}$. Light does not physically cut through a planet.*

Theorem 4.5 (A planet as a convex mirror; L06)**.** *For a sphere of radius R, in paraxial reprojection:*

- *the image of a distant source lies at depth $R/2$ inside the planet;*
- *an observer at height s sees their own image at depth $sR/(R+2s)$, with magnification $R/(R+2s)$.*

Proof. Mirror equation for a convex surface, $f = -R/2$. Battery: exact ray tracing gives $0.50000000R$ for a distant source and matches $sR/(R+2s)$ to 10^{-6} for $s = 0.5, 2, 10$. At aperture $0.5R$ the depth is $0.4226R$ (spherical aberration). A point image is a paraxial quotient; the full image is a caustic. □

5 What the image carries back

Theorem 5.1 (Logic of the mirror; L05)**.** *Take axes: x left–right, y vertical, z the mirror normal. Write M_{FB} for the front–back reflection the mirror performs, M_{LR} and M_{UD} for the left–right and up–down reflections, and $\mathrm{Rot}_\pi(\cdot)$ for a rotation by π about the stated axis. Then*

$$M_{\mathrm{FB}} = \mathrm{Rot}_\pi(\textit{vertical})\, M_{\mathrm{LR}} = \mathrm{Rot}_\pi(\textit{horizontal})\, M_{\mathrm{UD}}.$$

The mirror reverses only *front–back, along its normal. Whether one calls the image "left–right reversed" or "upside-down" depends on which rotation one uses to compare. This is a choice of complement, in exactly the sense of RG's complement-relative coupling. The invariants are chirality (*$\det = -1$*: a right glove images as a left glove) and photon helicity, which flips from $+1$ to -1.*

Theorem 5.2 (Moving mirror; L07)**.** *Reflect the wave four-vector in the mirror's rest frame and boost back. This reproduces Einstein's formulas [7] exactly:*

$$\frac{\nu_r}{\nu} = \frac{1 - 2\beta\cos\theta + \beta^2}{1 - \beta^2}, \qquad \cos\theta_r = -\frac{(1+\beta^2)\cos\theta - 2\beta}{1 - 2\beta\cos\theta + \beta^2}.$$

A receding mirror shifts by $(1-\beta)/(1+\beta)$. Your own image, as you approach a mirror at speed β, returns at $(1+\beta)/(1-\beta)$. The Galilean double shift $1-2\beta$ is wrong by $2\beta^2$.

Theorem 5.3 (Arbitrary mirror worldline, 1+1; L08)**.** *Consider a perfect mirror $x = z(t)$ with incident wave $f(t-x)$. The reflected wave is*

$$g(v) = -f\big(p(v)\big), \qquad p(v) = t(v) - z(t(v)), \qquad t + z(t) = v.$$

The mirror re-emits the incident wave re-projected along its own null map [8]. The local Doppler factor at the emission event is $p'(v) = (1 - \dot{z})/(1 + \dot{z})$.

For the uniformly accelerated mirror $z = \sqrt{a^2 + t^2} - a$ *one gets* $p = av/(a+v) \to a$. *Incident light later than* $u = a$ *never reaches the mirror, so the image* freezes *at a horizon.*

Corollary 5.4 (What the image carries)**.** *The image returns the observer's own field:*

- *CP-transformed: parity along the normal, and opposite helicity;*
- *delayed by the optical length;*
- *Doppler-shifted by the relative motion, or by* p' *in general;*
- *attenuated by the reflection coefficient that emission computes.*

6 Gravity: radar as emission and reception

Theorem 6.1 (Static gravity; L09, imported round trip)**.** *In a static spacetime, light follows the optical metric* h/N^2, *the* $v = c$ *member of the Jacobi pencil [2]. The echo of a static reflector is re-emission at the reflector. For Schwarzschild the round trip is*

$$T = 2N_0\Big[(r - r_0) + 2m \ln \frac{r - 2m}{r_0 - 2m}\Big],$$

and $B = \frac{c}{2} z T'$ *holds identically. Because the Killing frequency is conserved, your own image in a* static *mirror is never frequency-shifted: the ratio is exactly* 1, *while the delay is* T. *From* $r_0 = 10m$ *to* $30m$ *the proper distance is* $21.21m$, *the emitter-time half-delay is* $20.13m$, *and the optical length is* $22.51m$. *Radar distance is not proper distance.*

7 Bounce, image, emission: the RG typing

Theorem 7.1 (Correspondence and lock)**.** *For far-field protocols, the three latent representations produce one observation law and so form a CORRESPONDENCE:*

- *a ray with Fresnel coefficients (bounce);*
- *a CP-image emitter (image);*
- *self-consistent surface or volume emission (emission).*

The fibre is broken, and emission is LOCKED, by any of the following protocols:

(i) *fields behind or inside the reflector (extinction);*

(ii) *variation of the material, since emission computes* r, t, *absorption and dispersion from* χ, *while bounce must take them as inputs;*

(iii) *penetration depth;*

(iv) *mirror motion, where the null map* p *is an emission property of the worldline.*

The bounce is the stationary-phase limit of the emission sum. Equal angles (Heron, Fermat) are the stationary points of the path phase through the emitting surface.

Remark 7.2 (What is and is not new). That reflection is re-emission by the charges of the reflector is established physics: Ewald–Oseen, and Feynman's account in QED [6]. The RG contribution is the *typing*. The image is a quotient point; bounce, image and emission are a correspondence; emission is identified by named protocols. The paper also gives one projection formula whose reductions are exact and battery-checked.

8 Gaps of current physics, typed

Issue	Status	RG reading
Mechanism of reflection	closed	Ewald–Oseen and QED: emission. The "bounce" is a teaching shorthand, a stationary-phase limit.
Logic of the mirror (left–right)	closed	Front–back only; the attribution depends on the comparison rotation [13]. Chirality and helicity are the invariants.
Moving mirrors	closed (classical)	Einstein 1905; arbitrary worldlines through the null map p.
Photon "path"	open in interpretation	There is no standard position operator for the photon [11]. A photon trajectory is latent, not in the observable quotient; detection statistics are.
Momentum of light in media	largely resolved	Abraham and Minkowski momenta are two different natural modules, kinetic and canonical [12]; the task decides which applies.
Vacuum emission by mirrors	quantum, observed	Dynamical Casimir effect [10]. The Fulling–Davies flux $-\{p, v\}/24\pi$ [9] is a functional of the classical map p. Uniform acceleration (a Möbius p) emits nothing (L11). Classical RG supplies the input, not the quantum output.
Mirrors near horizons	closed (static)	The radar delay diverges logarithmically; the accelerated-mirror image freezes at $p \to a$.

There is no gap in the *physics* of the mirror. The gap is in its *description*: textbooks use bounce, image and emission interchangeably without saying that they are one observable law with three latent readings.

9 Ladder of the light path

Rung	Statement	Status	Witness
$\Lambda 0$	Declarations: geometry, Maxwell, media, observer protocol	declared	—
$\Lambda 1$	Projection formula: emission only, self-consistent	F (I: Maxwell)	L01, L04
$\Lambda 2$	Extinction behind and inside reflectors	F	L01, L04
$\Lambda 3$	Emission computes r, t, absorption from χ (Airy, second order)	F	L04
$\Lambda 4$	Planar mirror = CP image	F	L02
$\Lambda 5$	Image = quotient point; broken only behind the surface	F	L03
$\Lambda 6$	Finite penetration $\lambda/4\pi\kappa$; nothing passes through	F	L10
$\Lambda 7$	Convex planet: image inside at $R/2$; own image $sR/(R+2s)$	F	L06
$\Lambda 8$	Mirror reverses only the normal; LR attribution choice-dependent; chirality and helicity invariants	F	L05
$\Lambda 9$	Moving mirror = Lorentz-reflected image (Einstein)	F	L07
$\Lambda 10$	Arbitrary worldline: $g = -f \circ p$, Doppler p', horizon freezing	F	L08
$\Lambda 11$	Static gravity: optical metric round trip; own image unshifted; $B = \frac{c}{2} z T'$	F (H: static)	L09, [1]
$\Lambda 12$	Bounce/image/emission correspondence; emission locked by probe protocols	F	Thm 7.1

Rung	Statement	Status	Witness
Λ13	Vacuum emission: p is the complete classical input	I / boundary	L11
Λ14	Photon localization; non-static gravity; nonlinear media; real data	O	—

10 Battery and limits

`rg_light_attacks.py` (NumPy, SciPy, SymPy; master seed 20261025) runs 11 attacks: 10 locked and 1 boundary. Every canary is rejected by a *computed* criterion, never by a literal flag. The seven false statements are:

(1) the image's light passes through the mirror;

(2) single-pass emission gives the reflection coefficient;

(3) a mirror intrinsically swaps left and right;

(4) reflection preserves helicity;

(5) a planet's image sits at its centre;

(6) the Galilean Doppler shift is exact;

(7) radar distance equals proper distance.

The round-trip lock (`rg_roundtrip.py`, 23/23) is included and imported.

Limits. All data are analytic or synthetic, and the battery is not blind. The Ewald–Oseen check is one dimensional, at normal incidence. The planet is an ideal sphere in geometric optics; real planets are rough, and partly absorbing. The quantum statements are imported, not computed.

11 Non-claims

1. No new law of light: classical electrodynamics with the stated imports.
2. No claim that light physically passes through opaque matter.
3. No claim that current physics fails at the mirror; the paper corrects a representation, not a theory.
4. Novelty is claimed only for the RG typing and the unified projection formula with its batteries.

References

[1] M. Rada, *RG round-trip formula lock* (battery `rg_roundtrip.py`), 2026.

[2] M. Rada, *Resolution Geometry: The Duality of Gravity*, 2026.

[3] P. P. Ewald, Zur Begründung der Kristalloptik, *Ann. Phys.* 49 (1916) 1.

[4] C. W. Oseen, Über die Wechselwirkung zwischen zwei elektrischen Dipolen und über die Drehung der Polarisationsebene in Kristallen und Flüssigkeiten, *Ann. Phys.* 48 (1915) 1.

[5] M. Born, E. Wolf, *Principles of Optics*, 7th ed., Cambridge, 1999.

[6] R. P. Feynman, *QED: The Strange Theory of Light and Matter*, Princeton, 1985.

[7] A. Einstein, Zur Elektrodynamik bewegter Körper, *Ann. Phys.* 17 (1905) 891.

[8] G. T. Moore, Quantum theory of the electromagnetic field in a variable-length one-dimensional cavity, *J. Math. Phys.* 11 (1970) 2679.

[9] S. A. Fulling, P. C. W. Davies, Radiation from a moving mirror in two-dimensional space-time: conformal anomaly, *Proc. R. Soc. A* 348 (1976) 393.

[10] C. M. Wilson et al., Observation of the dynamical Casimir effect in a superconducting circuit, *Nature* 479 (2011) 376.

[11] T. D. Newton, E. P. Wigner, Localized states for elementary systems, *Rev. Mod. Phys.* 21 (1949) 400.

[12] S. M. Barnett, Resolution of the Abraham–Minkowski dilemma, *Phys. Rev. Lett.* 104 (2010) 070401.

[13] M. Gardner, *The Ambidextrous Universe*, Basic Books, 1964.

[14] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1999.